

Ayan Singha Roy

Bankura, West Bengal
+91 7319109362 — ayansingharoy7906@gmail.com
GitHub — LinkedIn

Professional Summary

B.Tech Computer Science & Engineering (AI & ML) student with hands-on experience in cybersecurity, Python development, machine learning, and network analysis. Experienced in building security-focused applications, AI-powered solutions, and real-world software projects using modern development tools and version control systems.

Education

Sanaka Educational Trust's Group of Institutions	2024 – Present
B.Tech in Computer Science & Engineering (AI & ML)	
Joypur High School	2024
Higher Secondary (PCMB) – 85.6%	
Changdoba High School	2022
Madhyamik – 76.42%	

Experience

Cybersecurity Intern — Prodigy InfoTech	2026
- Completed a cybersecurity internship focused on developing Python-based security and monitoring tools. - Designed and implemented encryption systems, activity monitoring applications, and network traffic analysis tools. - Applied concepts of cryptography, packet inspection, logging, and system monitoring in practical projects. - Utilized Git and GitHub for version control, documentation, and project management.	

Projects

LegalMetriX (Smart India Hackathon Project)
<i>Python, FastAPI, MongoDB, OCR, Gemini Vision API</i>
- Developed an AI-powered compliance verification system for packaged commodities under Legal Metrology Rules 2011. - Integrated OCR and Gemini Vision APIs to extract and validate product label information. - Built backend APIs using FastAPI and implemented MongoDB database integration. - Automated compliance analysis workflows for regulatory verification.
AgriGuard AI (Ongoing)
<i>Python, Machine Learning</i>
- Developing an AI-powered crop disease detection and plant health monitoring platform. - Applying machine learning and image analysis techniques for disease identification from leaf images. - Building recommendation workflows to assist farmers with treatment suggestions.
Network Packet Analyzer
<i>Python, Scapy</i>
- Built a packet analysis tool capable of capturing and inspecting live network traffic. - Classified and analyzed TCP, UDP, and ICMP packets for monitoring and troubleshooting. - Applied packet sniffing and protocol inspection techniques using Scapy.

- GitHub: github.com/sroyayan/network-packet-analyzer

Keyboard Activity Monitor

Python, Tkinter

- Developed a desktop-based keyboard activity monitoring application.
- Recorded keystroke events with timestamps and generated activity logs.
- Implemented an interactive graphical user interface using Tkinter.
- GitHub: github.com/sroyayan/keyboard_activity_monitor

Image Encryption Tool

Python, Pillow (PIL)

- Developed an image encryption and decryption application using pixel-level transformations.
- Implemented image security workflows for protecting visual data.
- GitHub: github.com/sroyayan/image-encryption-tool

Caesar Cipher Encryption Tool

Python

- Developed a cryptography application implementing Caesar Cipher encryption and decryption.
- Supported user-defined shift values and interactive text processing.
- Demonstrated foundational cryptographic concepts.
- GitHub: github.com/sroyayan/caesar-cipher

Technical Skills

Languages	Python, C, SQL
Libraries & Frameworks	FastAPI, NumPy, Pandas, Pillow (PIL), Tkinter, Scapy
Tools & Platforms	Git, GitHub, VS Code, MongoDB, Linux (Basics)
Core Concepts	Machine Learning, Cybersecurity, Networking, OOP, Data Structures & Algorithms

Certifications

- Introduction to GitHub – GitHub Skills
- Generative AI Fundamentals – Google Cloud Skills Boost

Languages

English, Hindi, Bengali